Ex. no: 13

27/8/26

SLIDING WINDOW PROTOCOL

AIM:

To implement the flow control at the data link layer using the sliding window protocol and simulate the flow of frames from one node to another.

SENDER CODE:

```
message = input ("Enter the message: ")
window_size = int (input ("Enter window size: "))
frames = [ ]
for i, ch in enumerate (message):
    frames. append ((i+1, ch))
print ("\n Frames created: ")
for no, data in frames:
    print (f" Frame {no} : {data}")
with open ("Sender - Buffet. txt", "w") as file =
    for no, data in frames:
        file. Write (f" {no} {data} \n")
print ("\n Frames stored in Sender - Buffet. txt")
base = 0
while base < len (frames):
    print ("\n Current window:")
    for i in range (base, min (base + window
                - size, len (frames))):
        print (f" [{frames [i] [0] : {frames [i] [1]}
                ]", end = " ")
try:
    with open ("Receiver - Buffet. txt", "r") as
                file:
        response = file. readline (). strip()
```

```
except file Not found Error:
    response = " "
if response.starts with ("ACK"):
    ack = int (response.split()[1])
    print (f"ACK {ack} received")
    base = ack
    open ("Receiver-Buffer.txt", "w").
                                close()
elif response.starts with ("NACK"):
    nack = int (response.split()[1])
    print (f "NACK {nack} received.")
    print (f "Retransmitting frame {nack}
                                ...")
    open ("Receiver-Buffer.txt", "w").
                                close()
else:
    print ("\n frames sent.")
    input ("Run receiver program & press
            ENTER to continue).
    print ("All frames transmitted")
```

RECEIVER CODE:

```
expected_frame = 1
while True:
  try:
    with open ("Sender-Buffer.txt", "r") as
                                file:
      frames = file.readlines()
except file Not found Error:
    print ("Sender-Buffer.txt not found")
    break.
```

```
if not frame:
    print ("No frames available")
    break.
print ("\n frames received from sender")
  for line in frames:
      parts = line.split()
if len(parts) != 2: continue
frame_no, data = int(parts[0]),
                 parts[1]
print(f"frame {frame_no}:{data}")

if frame_no == expected_frame:
    print(f"frame {frame_no}
                 received)
with open("Receiver_Buffer.txt","w")
    print(f"ACK {frame_no} sent");
else
    print f:"Wrong frame received");
break,
```

INPUT:

Enter the message = HELLO

Enter window size: 3.

OUTPUT :

Frames created :

Frame 1 : H

Frame 2 : E

Frame 3 : L

Frame 4 : L

Frame 5 : O

frames stored in Sender_Buffer.txt

current window :

[1 : H] [2 : E] [3 : L]

frames sent

Run receiver program and press ENTER to continue.

RECEIVER OUTPUT :

frames received from sender :

Frame 1 : H

Frame 1 received correctly.

ACK 1 sent

RESULT :

The sliding window protocol was successfully implemented using python and the data transmission and ACK process was verified successfully.